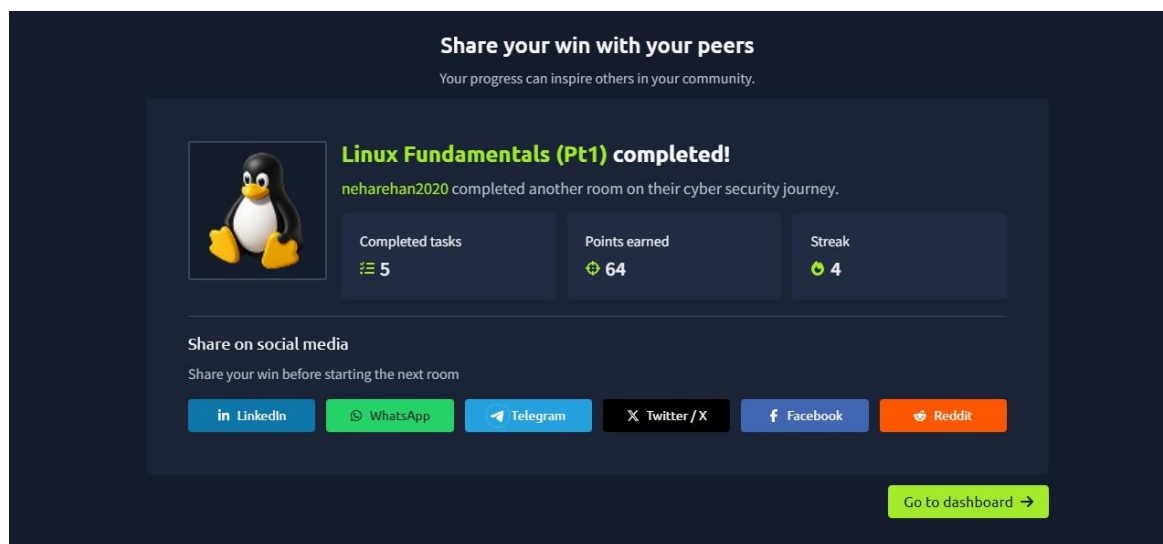
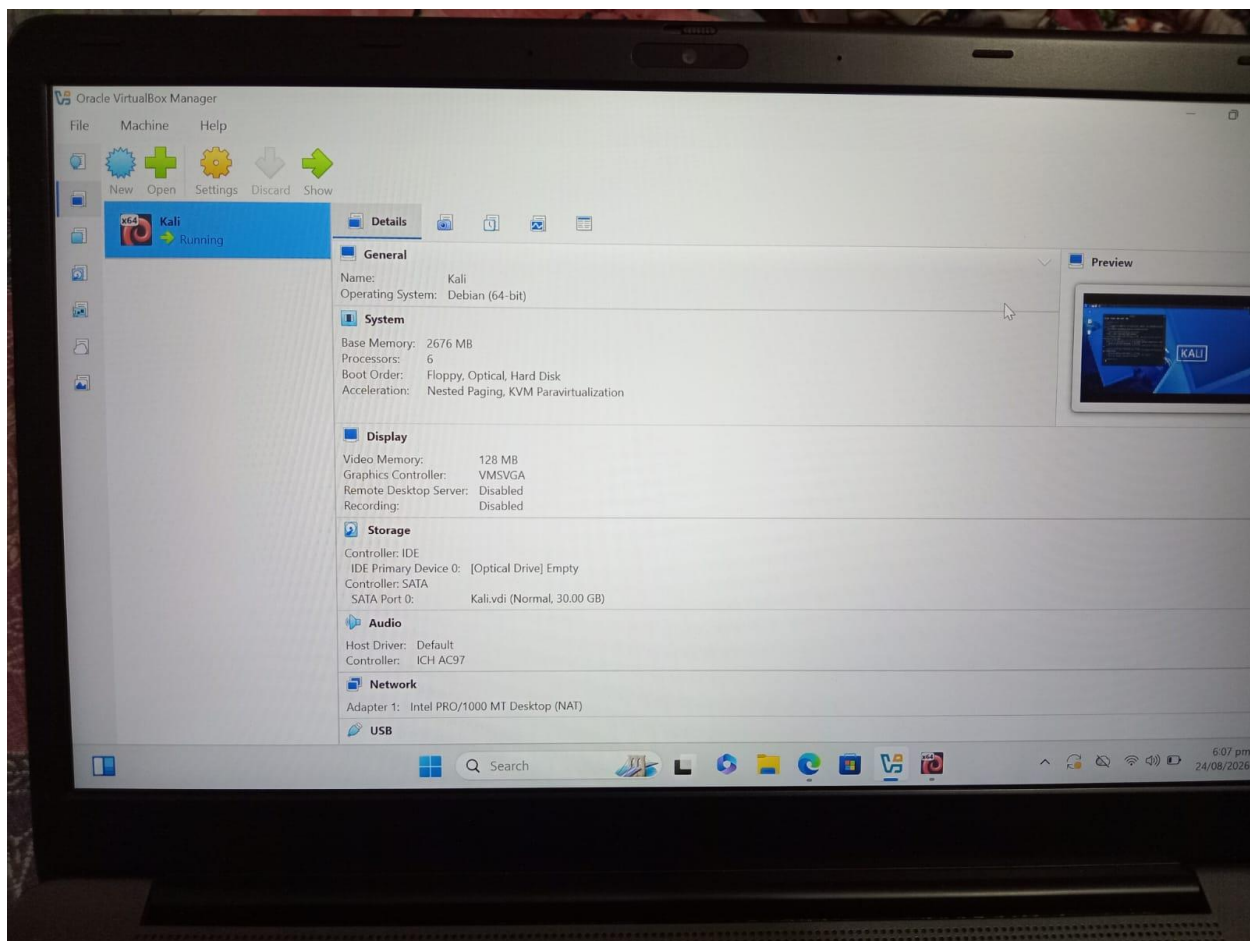


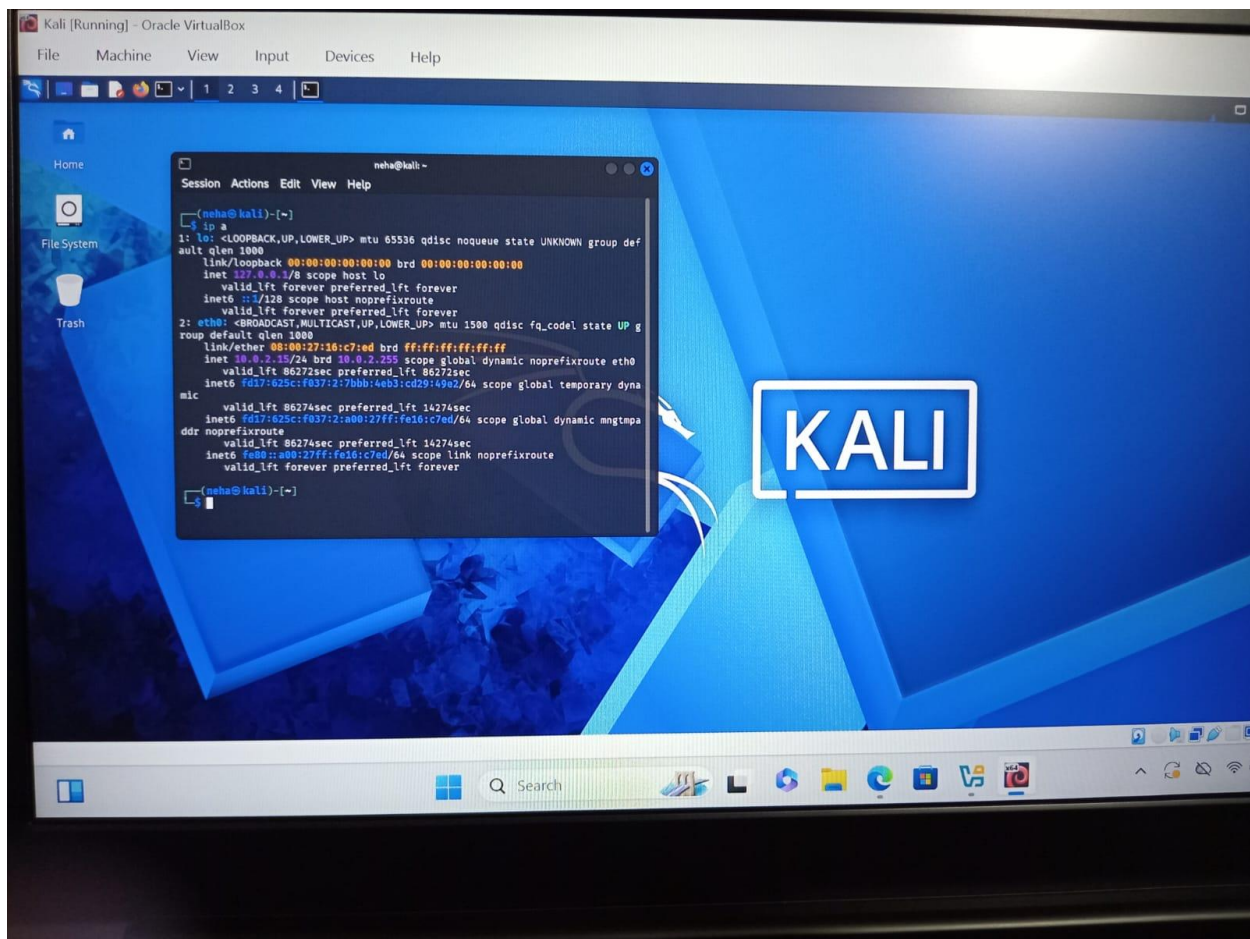
Week 1 · Task 3 – Lab Setup – VirtualBox + Kali Linux Installation

Key Definitions

- **VirtualBox:** A cross-platform, Type-2 hypervisor developed by Oracle. It allows a physical computer (host OS) to create and run multiple virtual machines (guest OSs) simultaneously without altering the host system.
- **Kali Linux:** An open-source, Debian-derived Linux distribution developed by Offensive Security. It is tailored specifically for ethical hacking, digital forensics, network security analysis, and penetration testing.
- **ip a Purpose:** Short for `ip address show`, this command is used in Linux terminal environments to display and configure network interfaces, showing active IP addresses (IPv4/IPv6), MAC addresses, and connection status for interfaces like `lo` (loopback) and `eth0` (Ethernet).







```
neha@kali: ~  
Session Actions Edit View Help  
  
(neha@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:16:c7:ed brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 86272sec preferred_lft 86272sec  
    inet6 fd17:625c:f037:2:7bbb:4eb3:cd29:49e2/64 scope global temporary dyna  
mic  
        valid_lft 86274sec preferred_lft 14274sec  
    inet6 fd17:625c:f037:2:a00:27ff:fe16:c7ed/64 scope global dynamic mngtmpa  
ddr noprefixroute  
        valid_lft 86274sec preferred_lft 14274sec  
    inet6 fe80::a00:27ff:fe16:c7ed/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(neha@kali)-[~]  
$
```